

Keys in Entity–Relationship Model

Concept of Keys

Definition

The values of an entity's attributes must be such that they **uniquely identify** the entity from the entity set. No two entities can have exactly the same values for *all* attributes. Keys are sets of attributes used for this unique identification, both for entities and relationships.

Hierarchy of Keys

Primary Key $\subseteq$ Candidate Key $\subseteq$ Superkey

Keys for Entity Sets

1. Superkey

Superkey

A set of one or more attributes that **uniquely** identifies an entity in the entity set. May contain extraneous attributes.

Example

Instructor: {ID}, {ID, name}, {all attributes} are superkeys. *name* alone is **not** a superkey.

2. Candidate Key

Candidate Key

A *minimal superkey* — a superkey with no proper subset that is also a superkey. Multiple candidate keys may exist.

Example

ID is a candidate key of **Instructor**. *course_id* is a candidate key of **Course**.

3. Primary Key

Primary Key

A candidate key chosen by the database designer as the **main** identifier for entities.

Keys for Relationship Sets

Superkey for Relationship Sets

The union of primary keys of the participating entity sets.

Example

Relationship set **advisor** has primary keys: `Instructor.ID` and `Student.ID` $\Rightarrow \{s_id, i_id\}$ is the superkey.

Note

Mapping cardinality affects the choice of primary key in relationship sets.

Primary Key Choice by Mapping Cardinality

Mapping Type	Primary Key Choice
Many-to-Many	Union of both entity primary keys
One-to-Many	Primary key of the Many side
Many-to-One	Primary key of the Many side
One-to-One	Primary key of either participating entity